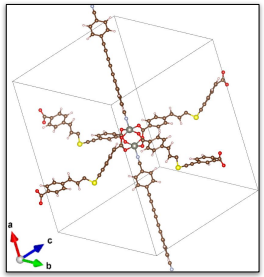


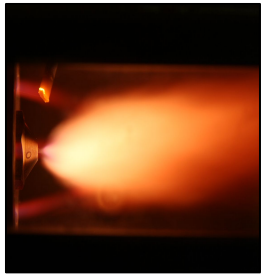
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Agentic Systems and DOE science

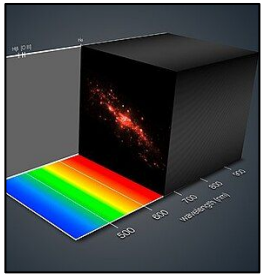
Agentic systems are being adopted across DOE



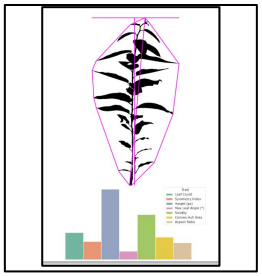
Materials Discovery



Combustion



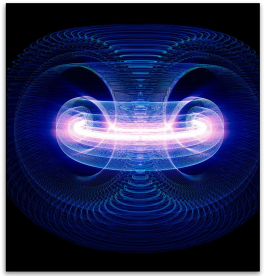
Astrophysics



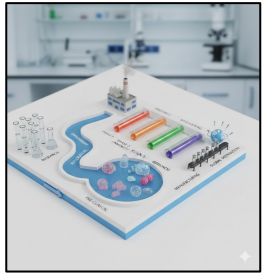
Plant Science



Engineering



Fusion



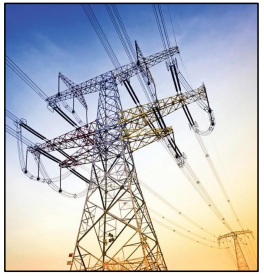
Biology



Lightsources



HPC Code



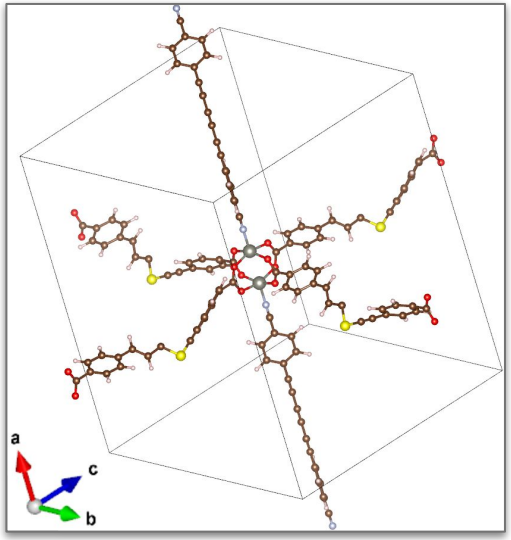
Power Grid

Searching enormous materials search spaces

Metal Organic Frameworks (MOF)

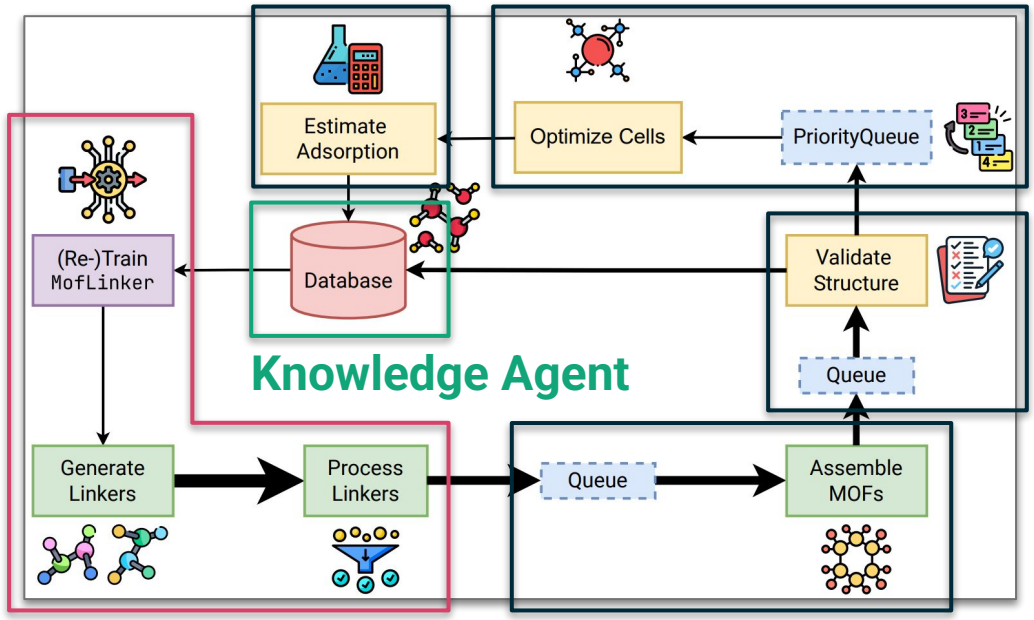
- Composed of organic molecules (ligands) and inorganic metals (nodes)
- The sponges of materials science!
- Porous structures that adsorb and store gases
- Topologies can be optimized for targeted gas storage → **Methane Capture**

How to efficiently discover MOFs with desirable properties for target applications?



Intractable search space of ligand, node, & geometry combinations

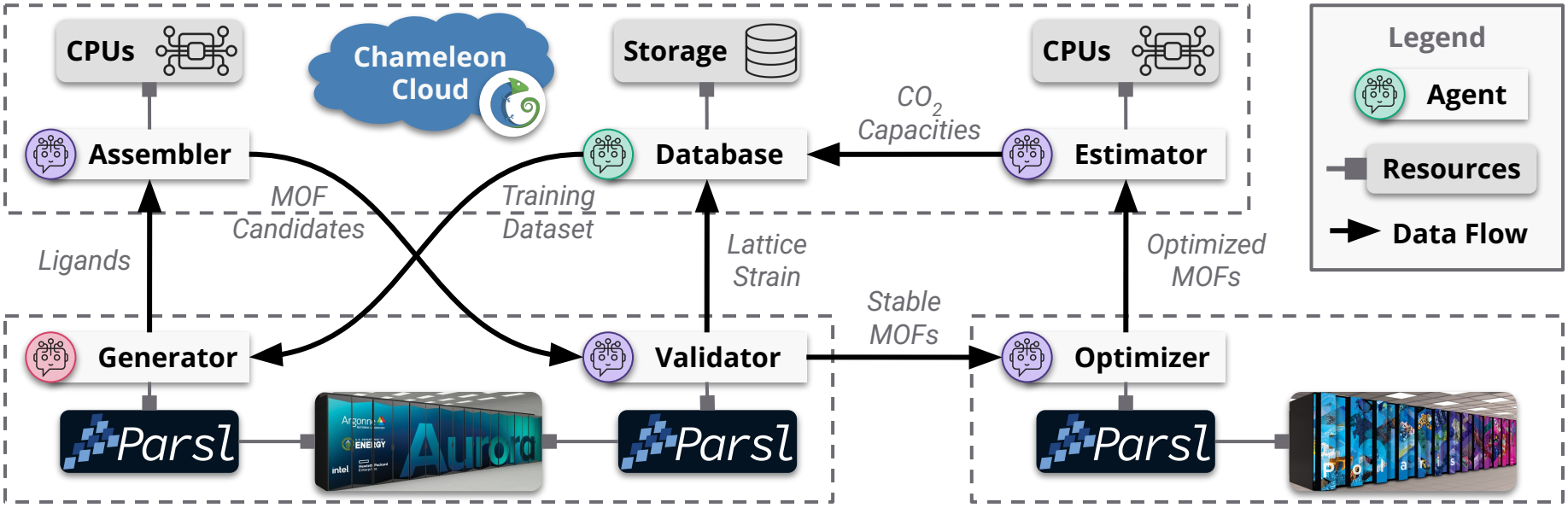
MOFA: Online learning + GenAI + Simulation



Computational Agents



MOFA realized with autonomous agents



Agents executed remotely via Globus Compute

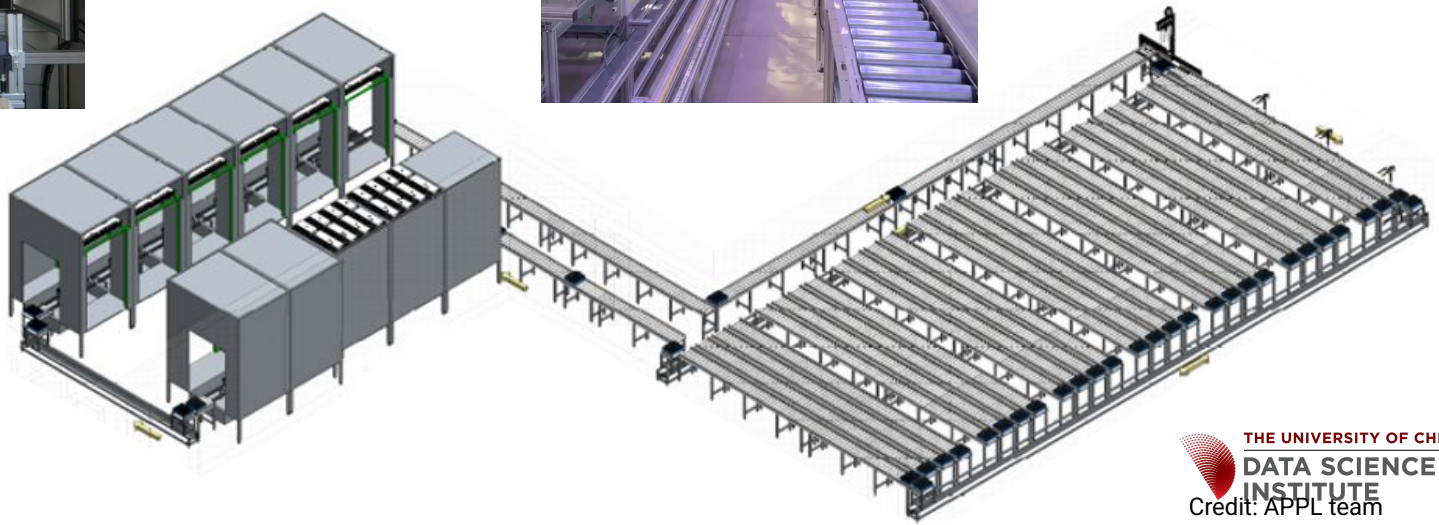
APPL is an automated 520-tray conveyor system that continually monitors plants using multimodal imaging.





Buffer

Imaging loop



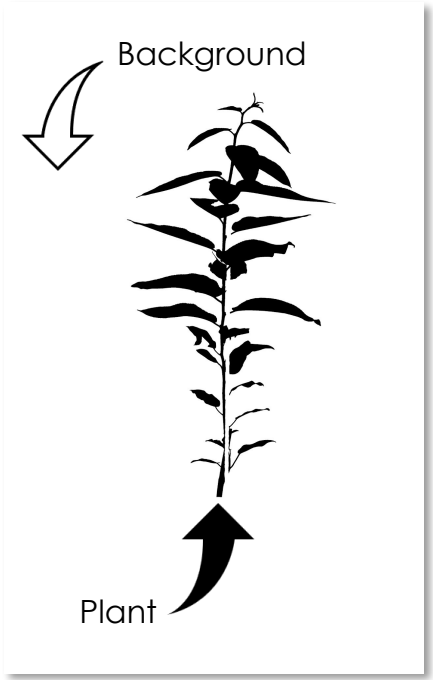
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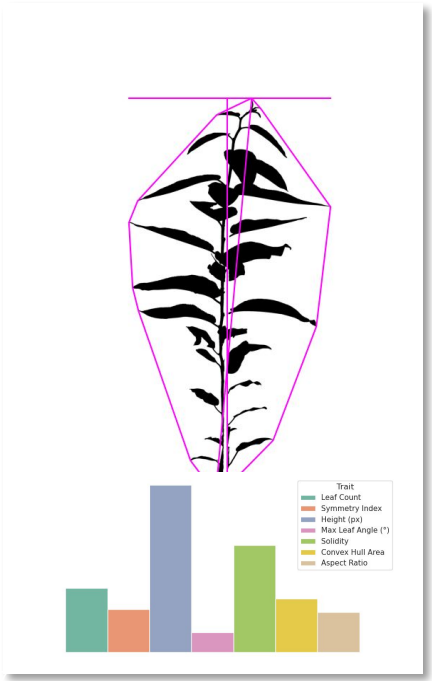
Credit: APPL team

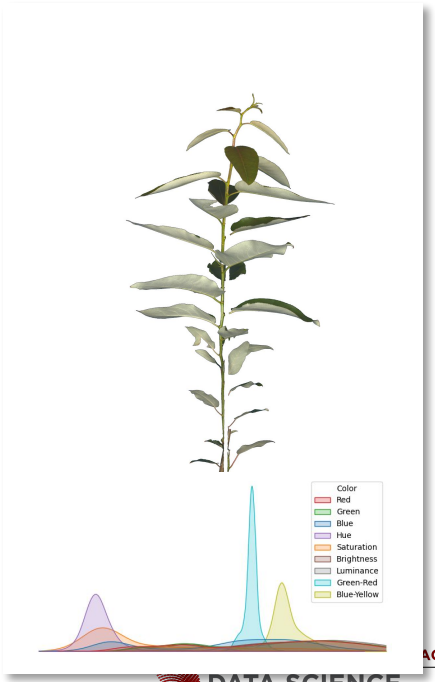
Accurate image segmentation enables prediction of complex plant phenotypes

Original → Segmentation → Morphology → Spectral traits









DATA SCIENCE INSTITUTE

Credit: APPL team



Daniel Rosendo



Renan Souza

Scientist

Genesis Mission
Platform Deployment for OPAL

Cloud

- **Model Services** *Model Catalog*
- **Intelligent Int.** *LiteLLM Inference*
- **i2 Auth** *Globus / Google*
- **Metadata Services** *Data lakehouse*
- **Co-pilot Agent** *Academy*
- **Chat UI** *Chainlit*
- **Provenance Service** *Flowcept*
- **Kubernetes cluster**
- aws

S3M

OLCF

- **ViT inference** *Parsl* — **Compute node**
- **Compute Agent** *Academy* — **Login node**
- **Lustre filesystem**
- FRONTIER

APPL





Plant images + experiment metadata

Platform

Legend

- APPL components
- Frontier
- AmSC services
- AWS services

Demo video
https://www.youtube.com/watch?v=7szA15ncDx0

Credit: APPL team

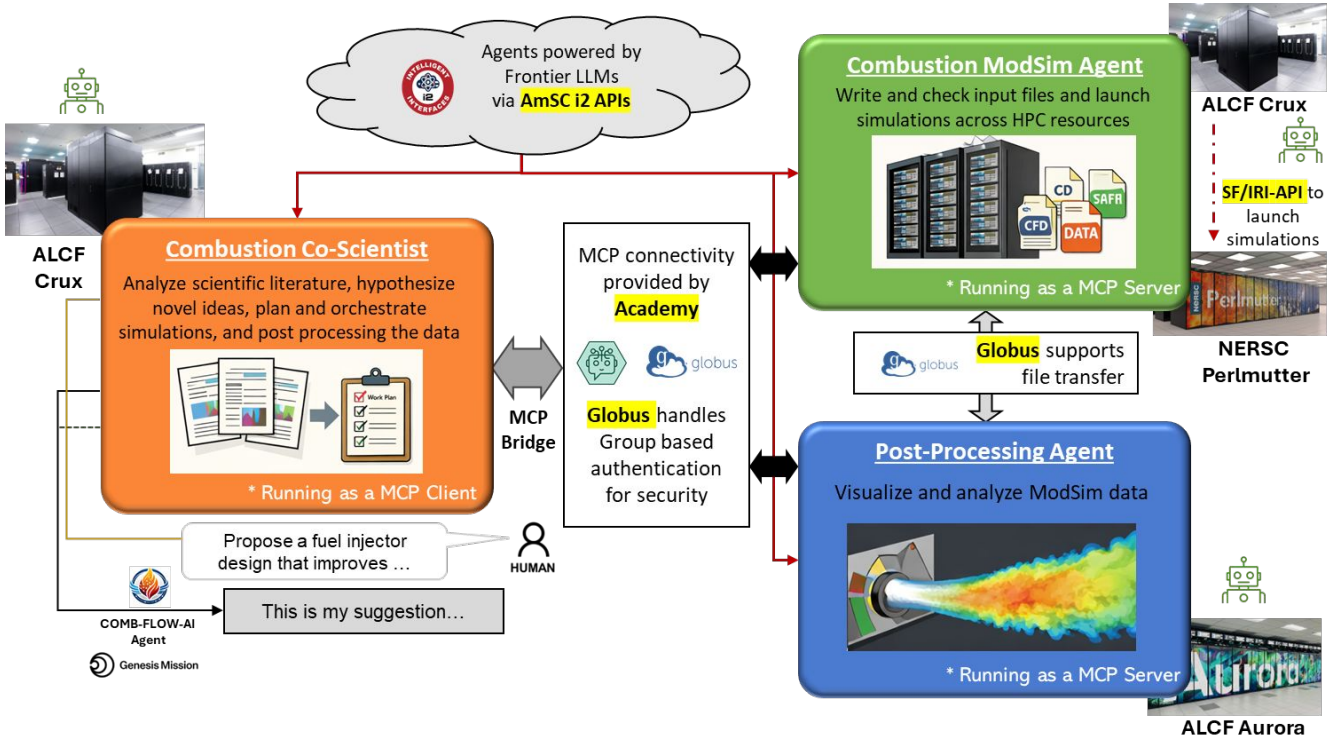
AISAC POWERS MANY DOMAIN APPLICATIONS

Use-Case : Genesis Combustion Model Team's Combustion Co-Scientist



Chandruchur Bhattacharya

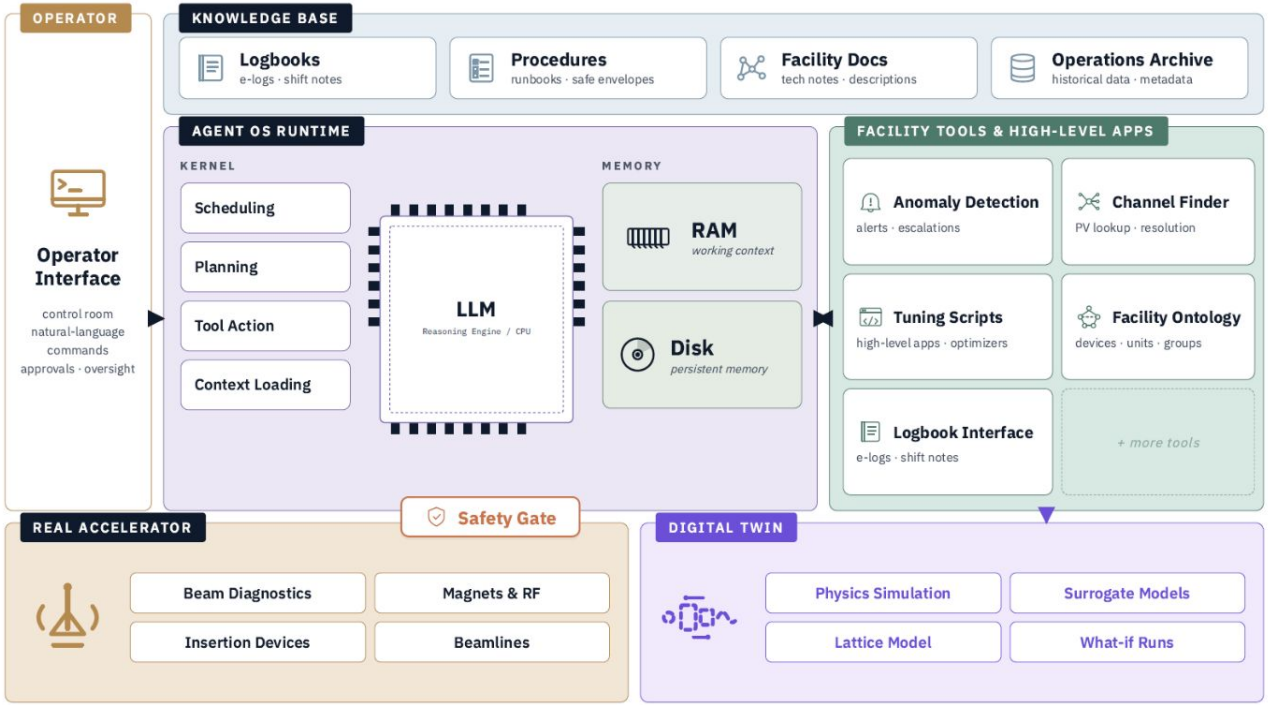
- **AISAC powers Combustion Co-scientist** that performs reasoning, knowledge synthesis, and user-interaction
- Combustion Co-scientist hands off tasks to other agents living on different HPC hardware via Academy
- Globus ensure authentication and security
- User can launch jobs on NERSC / ALCF from their laptops, running as server, or connected to a HPC-hosted instance
- Full COMB-FLOW-AI Agent system automates job launch, monitoring and post-processing
- Cross DOE-facility agent and task coordination



Chandrachur Bhattacharya, Martin Rieth, Kevin Griffin, Bruce Perry, Jean Sexton, Kiran Yalamanchi, Yixuan Sun, Sandeep Madireddy, Pei Zhang, Paul Laiu, Ann Almgren, Navin Mahto, Pinaki Pal, Jackie Chen, Shashank Yellapantula, Ramanan Sankaran, Sibendu Som

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Osprey — LLM as an Operating System for Accelerator Operations



An open framework for operating scientific instruments with agentic AI. The operator asks in plain language; the agent plans and acts on real hardware.

Osprey is the periphery that makes a coding agent work on large-scale scientific infrastructure — the connectors, facility knowledge, safety policy, and operator workflow a commercial agent lacks, with the agent itself replaceable.

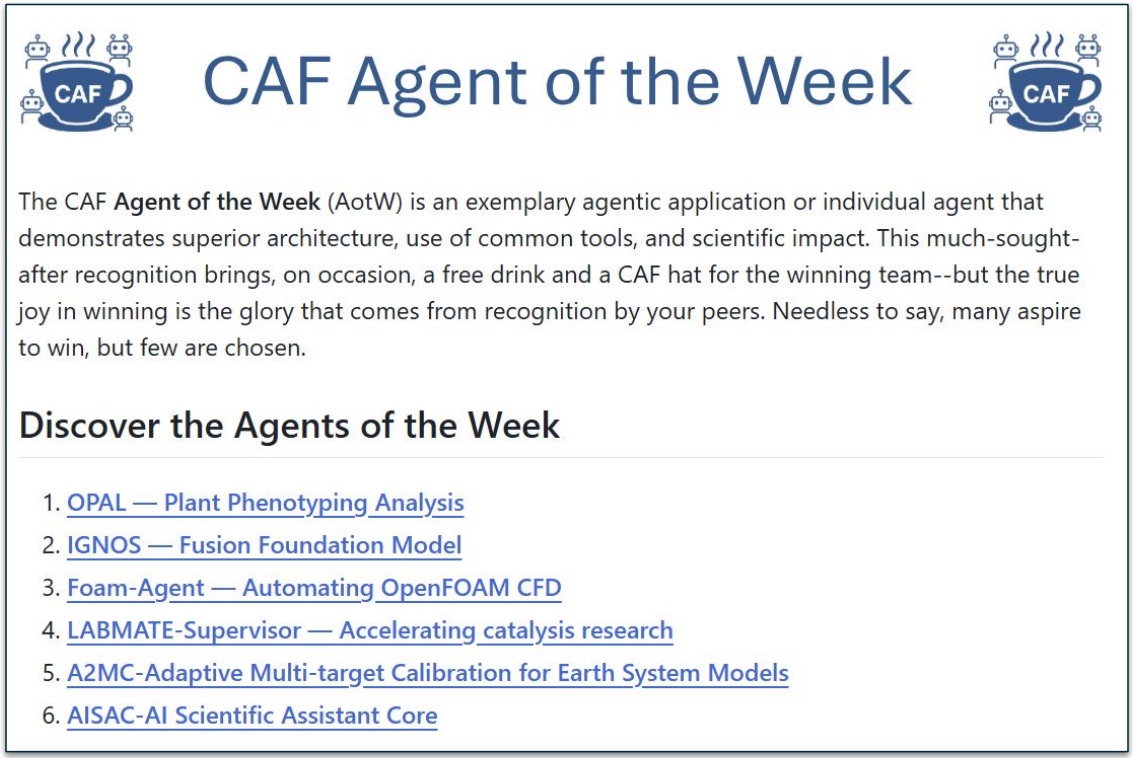
Safety is enforced around the agent, not by it — a layered chain above a hardware backstop it cannot disable.

Portable across facilities by configuration, not forks — the same harness drives machine tuning and logbook search.

Developed and demonstrated across DOE accelerator facilities — Berkeley Lab, SLAC, Argonne, Oak Ridge, Jefferson Lab, and Fermilab.

Credit: Thorsten Heller
https://arxiv.org/abs/2508.15066

Learn more about DOE agents



CAF Agent of the Week

The CAF **Agent of the Week** (AotW) is an exemplary agentic application or individual agent that demonstrates superior architecture, use of common tools, and scientific impact. This much-sought-after recognition brings, on occasion, a free drink and a CAF hat for the winning team--but the true joy in winning is the glory that comes from recognition by your peers. Needless to say, many aspire to win, but few are chosen.

Discover the Agents of the Week

1. OPAL — Plant Phenotyping Analysis
2. IGNOS — Fusion Foundation Model
3. Foam-Agent — Automating OpenFOAM CFD
4. LABMATE-Supervisor — Accelerating catalysis research
5. A2MC-Adaptive Multi-target Calibration for Earth System Models
6. AISAC-AI Scientific Assistant Core

https://github.com/AI-ModCon/BaseCAF_agent_of_the_week